

Group Project 4: Domain-Specific Access Control System

CS 428/528: Computer Security

Project Proposal Due Date : Apr-6

Project Due Date : Apr-17

Overview

In this project, you will work in small groups (**up to 5 students per group**) to design and implement a domain-specific access control system.

Each group will select a real-world domain and design an access control model using:

- Role-Based Access Control (RBAC)
- Attribute-Based Access Control (ABAC)
- Hybrid RBAC + ABAC

Your system should reflect real-world security requirements and policy enforcement.

Objective

The goal is to design and implement a practical access control system that demonstrates:

- Role assignment
- Permission assignment
- Access decision logic
- Policy enforcement
- Security constraints

Projects should demonstrate:

- Least privilege
- Separation of duties
- Role hierarchy (if applicable)
- Dynamic attributes (for ABAC)

Example Domains

Each team must select one domain (or request approval):

- Healthcare system
- Smart grid / energy system
- Cloud infrastructure
- University system
- Banking system

Project Proposal (Required)

Each team must submit a 1 page proposal including:

- Selected domain
- Roles definition
- Objects/resources definition
- Access rights
- Attributes (if using ABAC)
- Example policy rules
- Implementation plan

Project Requirements

Your implementation must include:

- Role definitions
- User assignment to roles
- Permission definitions
- Access decision function

Optional advanced features:

- Attribute-based policies
- Role hierarchy
- Separation of duties
- Logging and auditing
- Policy visualization

Implementation Options

You may implement using:

- Python
- Java
- Web application
- Database system
- Simulation

What to Submit

- Source code
- Project report (3–5 pages)
- Policy description
- Demo or presentation

Evaluation Criteria

Projects will be evaluated based on:

- Correctness of access control logic
- Realism of domain
- Security design
- Code quality
- Demonstration

Use of AI

AI tools are allowed for:

- Brainstorming, Design assistance, Implementation guidance

However:

- You must understand your implementation
- You must document AI usage